THE TIGHT-TILE LEMMA FOR THE $k^2 - 5$ FAMILY: STATEMENT AND PROOF

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Abstract. We give the full statement and proof of the Tight-Tile Lemma used in the companion paper to show that the measure μ_{11} of the $k^2 - 5$ family is valid on $[0, 11]^2$. The lemma concerns unit squares near *tight tiles*: axis-parallel unit squares next to a wall whose mass, as a limit at zero tilt, is exactly 1. The proof blows up the neighborhood of each tile with the tilt as scale variable, bounds the mass on each cell of the blown-up space from below by a concave function built from exact limit values and interval-certified derivative bounds, and checks this bound at the cell's vertices in exact rational arithmetic. We state every lemma used, describe the program `ttbox` that carries out the case analysis, list its soundness rules, and report the computation, the coverage check, controls and an independent review.

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1. Setting

1.1. **The measure.** $\mu = \mu_{11}$ is the measure on $[0,11]^2$ given by the file `sol_exact_box11.txt` (SHA-256 `d259fed5...adbd12099`, full hash in the release). It is the sum of

- 13,244 axis-parallel segments $\gamma_i = \{g_i + \tau(g_i' - g_i) : \tau \in [0,1]\}$, $g_i, g_i' \in \frac{1}{10}\mathbb{Z}^2$, each with weight $w_i > 0$ (the mass of the whole segment, spread uniformly by length; the smallest weight is $1.01 \cdot 10^{-10}$), and
- Lebesgue measure on $U = [\frac{19}{5}, \frac{36}{5}]^2$.

μ is invariant under the eight symmetries of $[0,11]^2$ (checked in exact arithmetic).

1.2. **Poses.** For a center $c \in \mathbb{R}^2$ and an angle θ, let $u = \tan(\theta/2)$, $N = 1 + u^2$ and

$$e_1 = \Big(\frac{1-u^2}{N},\ \frac{2u}{N}\Big), \qquad e_2 = \Big(-\frac{2u}{N},\ \frac{1-u^2}{N}\Big).$$

The pose is the closed unit square $P(c,u) = \{z \in \mathbb{R}^2 : |(z-c)\cdot e_1| \le \frac{1}{2},\ |(z-c)\cdot e_2| \le \frac{1}{2}\}$, and

$$\mu(P) = \sum_i w_i\, \ell_i(P) + |P \cap U|, \qquad \ell_i(P) = \big|\{\tau \in [0,1] : g_i + \tau(g_i' - g_i) \in P\}\big|.$$

1.3. **Tight tiles.** By symmetry we treat the left wall $x = 0$. The tile at depth $j \in \{1,2,3\}$ and height y is the axis-parallel unit square with center (d_0, y), $d_0 = j + \frac{1}{2}$. A pose near it is written with center $(d_0 + \delta,\ y)$ and $\tan(\theta/2) = \sigma u$, where $\sigma \in \{+1, -1\}$ is the *tilt sign* and $u > 0$.

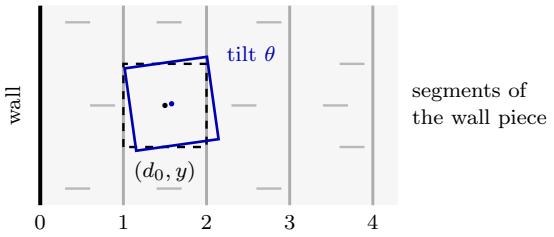


Figure 1. A tight tile at depth 1 (dashed) and a nearby pose with center $(d_0+\delta, y)$ and tilt θ (blue, tilt exaggerated). In the limit $\theta \to 0$ the tile's mass is exactly 1.

1.4. **Parameters.** $K = 4$, $M = 4$, $u_0 = 1/4000$, $\delta_{\max} = Ku_0 = 1/1000$, and the height ranges

$$I(3/2) = [1.4, 1.7] \cup [2.4, 2.7] \cup [3.3, 5.5],$$
$$I(5/2) = [1.4, 1.7] \cup [2.3, 2.7] \cup [3.3, 3.7] \cup [3.8, 5.5],$$
$$I(7/2) = [1.4, 1.7] \cup [2.3, 2.7] \cup [3.3, 3.7] \cup [4.1, 4.5].$$

On five further ranges (Table 1) we use $u_0 = 1/2000$ and $\delta_{\max} = 1/500$.

d_0	σ	ranges of y with $u_0 = 1/2000$
3/2	+1	$[4.2, 4.3]$, $[4.5, 4.6]$
3/2	−1	$[3.8, 3.9]$, $[5.4, 5.5]$
5/2	−1	$[4.4, 4.5]$

Table 1. Ranges certified with the larger tilt bound.

2. Statement

Theorem 2.1 (Tight-Tile Lemma)**.** *Let $d_0 \in \{3/2, 5/2, 7/2\}$, $\sigma \in \{+1, -1\}$ and $y \in I(d_0)$. Every pose with center $(d_0+\delta, y)$ and $\tan(\theta/2) = \sigma u$, where $0 < u \le u_0$ and $|\delta| \le \delta_{\max}$, has $\mu(P) \ge 1$. The same holds with $u_0 = 1/2000$ and $\delta_{\max} = 1/500$ for the ranges of Table 1, each extended by $1/500$ at both ends.*

Remark 2.2 (how the lemma is used). The interval checker `zmx2` certifies all poses of $[0,11]^2$ except the sub-boxes it leaves in 76 root boxes. One of them (root 9632) is closed by the exact checker `qx2_zm.py`. The other 75 are neighborhoods of tight tiles, and Section 9 shows that every sub-box left there consists of poses covered by Theorem 2.1 or of untilted poses ($u = 0$), which are certified separately.

3. Exact chords

Lemma 3.1 (clip formula)**.** *For a segment $\gamma = \{g + \tau(g' - g)\}$, each of the four half-planes of $P(c,u)$ reads $\tau A \le B$ with*

$$A = s\,(g' - g)\cdot Ne_i, \qquad B = \tfrac{N}{2} - s\,(g - c)\cdot Ne_i, \qquad i \in \{1,2\},\ s \in \{+1,-1\}.$$

Let $\mathcal{B}^+$ consist of 1 and the ratios B/A with $A > 0$, and $\mathcal{B}^-$ of 0 and the ratios with $A < 0$. If some constraint has $A = 0$ and $B < 0$, then $\ell = 0$; constraints with $A = 0 \le B$ hold for all τ. Otherwise

$$\ell(P) = \max\bigl(0,\ \min \mathcal{B}^+ - \max \mathcal{B}^-\bigr).$$

Proof. $|(z-c)\cdot e_i| \le \frac{1}{2}$ is the pair of inequalities $s(z-c)\cdot e_i \le \frac{1}{2}$. Substituting $z = g + \tau(g'-g)$ and multiplying by $N > 0$ gives $\tau A \le B$. The set of $\tau \in [0,1]$ satisfying all four is the interval $[\max \mathcal{B}^-, \min \mathcal{B}^+]$ (empty if the bounds cross). □

With the factor N, the quantities A and B are polynomials in u and c with rational coefficients.

4. Charts

Throughout, the scale variable is the tilt: $h = u$. In each chart the pose has

$$c = c_0(p,q) + h\,c_1(p,q,h), \qquad \tan(\theta/2) = \sigma h,$$

with c_0 affine and c_1 polynomial in the *face coordinates* (p,q), so A and B are polynomials in (p,q,h) with rational coefficients.

Lemma 4.1 (division)**.** *Let h^m be the largest power of h dividing both A and B. On the region $h > 0$, the constraint $\tau A \le B$ is equivalent to $\tau(A/h^m) \le B/h^m$.*

Proof. $h^m > 0$. □

4.1. **Charts near a base line.** Let $y_b \in \frac{1}{10}\mathbb{Z}$ and $t = y - y_b$. The neighborhood of the line is $\mathcal{V}(y_b) = \{0 < u \le u_0,\ |t| \le Ku_0,\ |\delta| \le \delta_{\max}\}$.

chart	poses	face coordinates (p,q)	slab
C_u	$\lvert\delta\rvert, \lvert t\rvert \le Ku$	$(\delta/u,\ t/u) \in [-K,K]^2$	$0 \le h \le u_0$
$W_t^{\pm}$	$\lvert\delta\rvert \le Ku < \lvert t\rvert,\ \pm t > 0$	$(\delta/u,\ \lvert t\rvert) \in [-K,K]\times[0,Ku_0]$	$h \le \lvert t\rvert/K$
$W_d^{\pm}$	$\lvert t\rvert \le Ku < \lvert\delta\rvert,\ \pm\delta > 0$	$(t/u,\ \lvert\delta\rvert) \in [-K,K]\times[0,\delta_{\max}]$	$h \le \lvert\delta\rvert/K$
$W_{td}^{\pm\pm}$	$Ku < \lvert\delta\rvert,\ Ku < \lvert t\rvert$	$(\lvert t\rvert, \lvert\delta\rvert)$, triangles $\lvert t\rvert \le \lvert\delta\rvert$ or $\lvert\delta\rvert \le \lvert t\rvert$	$h \le \min(\lvert t\rvert, \lvert\delta\rvert)/K$

Table 2. Charts near the base line y_b.

4.2. **Strip charts.** For $y \in [y_b + Ku_0,\ y_b + \frac{1}{10} - Ku_0]$: Σ_u with face $(\delta/u,\ y)$ for $|\delta| \le Ku$ and slab $h \le u_0$; and $\Sigma_d^{\pm}$ with face $(y,\ |\delta|)$ for $\pm\delta > Ku$ and slab $h \le |\delta|/K$.

Lemma 4.2 (the charts cover)**.** *Every pose of Theorem 2.1 lies in one of these charts, with face coordinates in the stated ranges and $h = u$ in the stated slab.*

Proof. Each $I(d_0)$, and each range of Table 1, is a finite union of intervals $[y_1, y_2]$ with $y_1, y_2 \in \frac{1}{10}\mathbb{Z}$. Each y in such an interval satisfies $|y - y_b| \le Ku_0$ for a line $y_b \in \frac{1}{10}\mathbb{Z} \cap [y_1, y_2]$, or lies in a strip between consecutive lines of the interval. Near a line, $|t| \le Ku_0$ and $|\delta| \le \delta_{\max} = Ku_0$, and exactly one of the four cases of Table 2 holds. In C_u, $(p, q) = (\delta/u, t/u) \in [-K, K]^2$ and $h = u \le u_0$. In $W_t^{\pm}$, $p = \delta/u \in [-K, K]$, $q = |t| \le Ku_0$ and $h = u < |t|/K$; $W_d^{\pm}$ is symmetric. In W_{td}, $h < \min(|t|, |\delta|)/K$. In a strip the same split by $|\delta| \le Ku$ or $> Ku$ gives Σ_u or $\Sigma_d^{\pm}$. For the ranges of Table 1, the charts of the two end lines also cover heights within $Ku_0 = 1/500$ beyond the range, which gives the extension. □

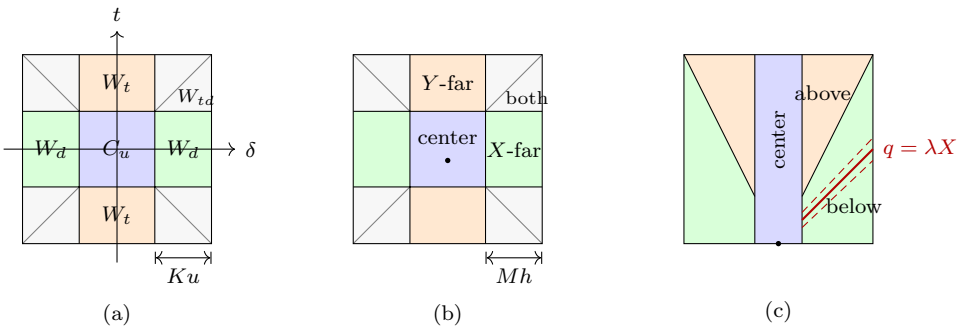


FIGURE 2. (a) The charts of Table 2 in the (δ, t) plane at a fixed tilt u. (b) The zones around a corner point of C_u (Section 6), at a fixed h. (c) The zones around a corner point of a wedge chart, and the near zone (dashed) around a ray $q = \lambda X$.

A chart whose slab is $h \le \chi\,\xi$ for a face coordinate $\xi \ge 0$ and a constant $\chi > 0$ is a *wedge chart* with wedge variable ξ.

5. The bound on a cell

Definition 5.1. A *cell* is $Z = \{(p, q, h) : (p, q) \in \Pi,\ \mathrm{low}(p, q) \le h \le \mathrm{high}(p, q)\}$, where Π is a convex polygon with rational vertices and $0 \le \mathrm{low} \le \mathrm{high}$ are affine. Its *box* is $\mathrm{Bx}(Z) = \mathrm{bbox}(\Pi) \times [0, \max_\Pi \mathrm{high}]$.

Lemma 5.2 (vertices)**.** *Z is a convex polytope whose vertices are among the points $(v, \mathrm{low}(v))$ and $(v, \mathrm{high}(v))$, v a vertex of Π.*

Proof. Z is an intersection of half-spaces. A point (z, h) of Z with $\mathrm{low}(z) < h < \mathrm{high}(z)$ is interior to a vertical segment in Z, and a point with z not a vertex of Π is interior to a segment in Π along which low (or high) is affine; neither is a vertex. □

5.1. **Classifying constraints.** All enclosures below are computed in binary64 interval arithmetic with outward rounding (Section 7), on $\mathrm{Bx}(Z)$. For each constraint $\tau A \le B$ (after Lemma 4.1) one of the following is established; otherwise the cell is refused and split.

- *Regular*: A has constant sign on $\mathrm{Bx}(Z)$, so B/A is C^1 there. In addition $A(p, q, 0)$ is a nonzero constant and $B(p, q, 0)$ is affine in (p, q).
- *Inactive*: $B \ge \max(0, A)$ on $\mathrm{Bx}(Z)$; then $\tau A \le B$ for all $\tau \in [0, 1]$.

- *Empty*: $B < \min(0, A)$ on $\mathrm{Bx}(Z)$; then $\tau A > B$ for all $\tau \in [0,1]$, so the chord is 0.
- *Wedge test* (wedge charts, when the two previous tests are inconclusive). Substitute $h = \eta\,\xi$ with $\eta \in [0,\chi]$, which is exact on the cell, divide B and $B - A$ by the largest power of ξ (positive), and decide inactive or empty from the signs of both. Near the apex of a wedge, one of two further exact substitutions is made: $\xi = \zeta\,\xi'$ with $\zeta \in [0, \zeta_{\max}]$ on a region where $\xi \le \zeta_{\max}\xi'$ (the chart provides $\zeta_{\max}$, by default 1), or $\xi' = \zeta\,\xi$ with $\zeta \in [-1,1]$ on a region where $|\xi'| \le \xi$.

Lemma 5.3 (bisection of auxiliary variables)**.** *If an interval test on the box of (p,q,η,ζ) is inconclusive, the checker bisects the widest of the ranges of η,ζ (up to 8 levels) and accepts a verdict only if it is the same, inactive or empty, on every piece. This verdict holds on the whole box.*

Proof. The pieces are closed intervals covering the range, so every point lies in a piece where the verdict holds. □

5.2. **Limit values and mean-value bounds.** For a regular constraint the *limit value* $b(0) = B(p,q,0)/A(p,q,0)$ is affine on Π. For each segment, write b_r^+ for the ratios B/A of its regular constraints with $A > 0$ together with the constant bound 1, and b_r^- for those with $A < 0$ together with the constant bound 0; a constant bound has derivative 0, inactive constraints are dropped, and a segment with an empty constraint contributes 0 on the whole cell. The checker splits Π exactly along affine lines until, on all of Π (checked at the vertices of Π, which suffices for affine functions), the minimizing upper bound $b_{r_*}^+(0)$, the maximizing lower bound $b_{r_*}^-(0)$ and the sign of $f(0) = b_{r_*}^+(0) - b_{r_*}^-(0)$ are fixed.

Let α_r and β_r be outward-rounded enclosures with

$$\alpha_r \le \inf_{\mathrm{Bx}(Z)} \partial_h b_r^+, \qquad \beta_r \ge \sup_{\mathrm{Bx}(Z)} \partial_h b_r^-, \qquad \partial_h \frac{B}{A} = \frac{A\,\partial_h B - B\,\partial_h A}{A^2}.$$

Lemma 5.4 (mean value)**.** *For every $(p,q,h) \in Z$: $b_r^+(h) \ge b_r^+(0) + h\,\alpha_r$ and $b_r^-(h) \le b_r^-(0) + h\,\beta_r$.*

Proof. The vertical segment $\{(p,q)\} \times [0,h]$ lies in $\mathrm{Bx}(Z)$, where $b_r^\pm$ is C^1; apply the mean value theorem. □

Lemma 5.5 (concave chord bound)**.** *Let* $\mathrm{gap}_r^+ = b_r^+(0) - b_{r_*}^+(0) \ge 0$, $\mathrm{gap}_r^- = b_{r_*}^-(0) - b_r^-(0) \ge 0$ *and*

$$\varphi = \min_r(\mathrm{gap}_r^+ + h\,\alpha_r) + \min_r(\mathrm{gap}_r^- - h\,\beta_r).$$

Then φ is concave in (p,q,h), and with $f(h) = \min \mathcal{B}^+ - \max \mathcal{B}^-$ and $\ell = \max(0,f)$,

$$\ell(h) - \ell(0) \ge \max(\mathrm{opt}_0, \mathrm{opt}_1), \quad \begin{cases} \mathrm{opt}_0 = \varphi,\ \mathrm{opt}_1 = -f(0) & \text{if } f(0) \ge 0 \text{ on } \Pi, \\ \mathrm{opt}_0 = f(0) + \varphi,\ \mathrm{opt}_1 = 0 & \text{if } f(0) \le 0 \text{ on } \Pi. \end{cases}$$

For $\kappa \in [0,1]$, $\kappa\,\mathrm{opt}_1 + (1-\kappa)\mathrm{opt}_0$ is concave and is a lower bound for $\ell(h) - \ell(0)$.

Proof. The gaps are affine, so φ is a sum of minima of affine functions. By Lemma 5.4, $\min \mathcal{B}^+(h) \ge b_{r_*}^+(0) + \min_r(\mathrm{gap}_r^+ + h\alpha_r)$ and $\max \mathcal{B}^-(h) \le b_{r_*}^-(0) - \min_r(\mathrm{gap}_r^- - h\beta_r)$, so $f(h) \ge f(0) + \varphi$. If $f(0) \ge 0$ then $\ell(0) = f(0)$, so $\ell(h) - \ell(0) \ge \varphi$ and $\ge -f(0)$. If $f(0) \le 0$ then $\ell(0) = 0$, $\ell(h) \ge f(0) + \varphi$ and $\ell(h) \ge 0$. A convex combination of two lower bounds is at most their maximum; a convex combination of concave functions is concave. □

5.3. **The Lebesgue term.** Only the depth-7/2 tiles reach U. Let $a(h) = |P \cap U|$.

- *Limit value.* At $h = 0$ the pose is axis-parallel and $a(0) = A_x(p,q)A_y(p,q)$, where A_x, A_y are the lengths of the projections of $P \cap U$, which are clamped affine functions of (p,q). The polygon is split at their kinks, so both are affine on Π. If both vary, the McCormick bound $A_xA_y \geq A_x^{\min}A_y + A_xA_y^{\min} - A_x^{\min}A_y^{\min}$ is used; it is affine and valid since $(A_x - A_x^{\min})(A_y - A_y^{\min}) \geq 0$.
- *Derivative.* a is Lipschitz in h, and for almost every h, by the transport formula with U fixed, $a'(h) = \int_{\partial P \cap U} v \cdot n \, ds$, where v is the velocity of the boundary point and n the outer normal. On an edge of P with parameter $s \in [-\frac{1}{2}, \frac{1}{2}]$, $v \cdot n = c' \cdot n + \theta' \, \epsilon \, s$ with $\epsilon = \pm 1$ fixed by the edge. The part of the edge in U is an interval $[s_{lo}, s_{hi}]$ obtained by clipping the edge against the four lines of U exactly as in Lemma 3.1 (including Lemma 4.1 and the wedge test). The integral equals $(s_{hi} - s_{lo})\, c' \cdot n + \theta'\epsilon \, (s_{hi}^2 - s_{lo}^2)/2$ and is enclosed by intervals; if the edge may miss U, the enclosure is widened to contain 0. The sum over the four edges gives a lower bound α_U for a' on $\mathrm{Bx}(Z)$, and $a(h) \geq a(0) + h\,\alpha_U$.

Proposition 5.6 (cell certificate)**.** *Let $G(p,q)$ be the exact affine limit value of $\mu(P) - 1$ on Π (with the McCormick bound for the Lebesgue part), let $\kappa_i \in [0,1]$ be rational, one per segment, and*

$$T = G + \sum_i w_i\big(\kappa_i \mathrm{opt}_{1,i} + (1 - \kappa_i)\mathrm{opt}_{0,i}\big) + h\,\alpha_U.$$

Then $\mu(P) - 1 \geq T$ on Z and T is concave. Hence if $T \geq 0$ at the vertices of Z, evaluated in exact rational arithmetic, then $\mu(P) \geq 1$ on Z.

Proof. $\mu(P) - 1 = G + \sum_i w_i(\ell_i(h) - \ell_i(0)) + (a(h) - a(0))$ up to the McCormick step, which only lowers G; apply Lemma 5.5 to each segment and the derivative bound to a. Segments that do not meet any pose of the cell contribute 0 and are omitted (Section 7). T is a sum of concave functions, and a concave function on a polytope attains its minimum at a vertex (Lemma 5.2). □

The κ_i are proposed by a floating-point linear program and then used as exact rationals; the linear program plays no role in soundness. A refused cell is split by bisecting Π, or along a *kink plane* $h = -f(0)/\dot{f}$, where $\dot{f}$ is the slope in h of a segment's bound, so that one branch of the maximum in Lemma 5.5 dominates on each side. Splitting stops at side length 2^{-30}; a cell still refused there is reported as a failure, never discarded.

6. Corner points

At isolated face points two events coincide, and the cells touching the point are refused at every scale. The program groups the refused cells of a chart by a nearby point with small denominators and re-covers a box around it by a second blow-up with scale h. Each of the three constructions below covers the set it replaces (Lemmas 6.1–6.3), and every zone is again a chart in the sense of Section 4, certified by Proposition 5.6. Zones may contain more poses than needed; this is harmless, since the pose formulas are polynomial identities and the bound is proved on each zone as a whole.

6.1. **Corners of C_u.** A refused cell of C_u is assigned to a point (p^*, q^*) with denominators at most 1000: a vertex of the cell, or a point within 2^{-20} of its centroid. Let ρ be the smallest power of 2 (at least 2^{-30}) such that the box $\Omega = [p^* \pm \rho] \times [q^* \pm \rho]$ contains all cells assigned to the point; $\rho \leq M u_0$ is required. With $X = |p - p^*|$ and $Y = |q - q^*|$:

zone	poses	face coordinates	slab
center	$X, Y \le Mh$	$(\bar p, \bar q) \in [-M, M]^2$, $(p, q) = (p^*, q^*) + h(\bar p, \bar q)$	$h \le u_0$
X-far$^{\pm}$	$X > Mh \ge Y$	$(X, \bar q)$, $p = p^* \pm X$, $q = q^* + h\bar q$	$h \le X/M$
Y-far$^{\pm}$	$Y > Mh \ge X$	$(\bar p, Y)$, $p = p^* + h\bar p$, $q = q^* \pm Y$	$h \le Y/M$
both far$^{\pm\pm}$	$X, Y > Mh$	(X, Y), triangles $X \le Y$ or $Y \le X$	$h \le \min(X, Y)/M$

Lemma 6.1. *These zones cover $\Omega \times [0, u_0]$.*

Proof. Each pose satisfies exactly one of $X \le Mh$, $X > Mh$ and one of $Y \le Mh$, $Y > Mh$. In the center zone $(\bar p, \bar q) \in [-M, M]^2$ and $h \le u_0$; in the far zones the defining strict inequalities are the slab bounds, which are at most $\rho/M \le u_0$. □

6.2. **Corners of wedge charts.** In a wedge chart with slab $h \le q/K$ the refused cells cluster at points $(p^*, 0)$ of the edge $q = 0$. A cell is assigned to p^* (denominator at most 1000) if its centroid is within 2^{-20} of $(p^*, 0)$; ρ is the smallest power of 2 for which $\Omega = [p^* \pm \rho] \times [0, \rho]$ contains the assigned cells. With $X = |p - p^*|$ and $M = K$:

zone	poses	face coordinates	slab
center	$X \le Mh$	$(\bar p, q) \in [-M, M] \times [0, \rho]$, $p = p^* + h\bar p$	$h \le q/K$
far$^{\pm}$, above	$X > Mh$, $q \ge X$	(X, q), triangle $q \ge X$	$h \le X/M$
far$^{\pm}$, below	$X > Mh$, $q \le X$	(X, q), triangle $q \le X$	$h \le q/K$

Lemma 6.2. *These zones cover $\Omega \times \{0 \le h \le q/K\}$.*

Proof. If $X \le Mh$ the pose is in the center zone with $\bar p = (p - p^*)/h \in [-M, M]$. Otherwise, if $q \ge X$ then $h < X/M$, and if $q \le X$ the original slab is kept. □

6.3. **Rays.** A far-below zone can in turn refuse cells along a ray $q = \lambda X$ through its apex. The program takes for λ the rational with denominator at most 20 in $(0, 1)$ whose ray meets the most refused cells (for a convex cell with $X > 0$ the values of q/X form the interval between their extremes at the vertices), and re-covers the whole zone. With $\Delta = q - \lambda X$, $M' = 1$, $\Delta_c = M'\lambda/(K - M')$ and $E_c = M'\lambda/(K + M')$:

zone	poses	face coordinates	slab
near	$\lvert\Delta\rvert \le M'h$	$(X, \bar z)$, $q = \lambda X + h\bar z$, $\lvert\bar z\rvert \le M'$	$h \le \lambda X/(K - M')$
above 1	$\Delta > M'h$, $\Delta \le \Delta_c X$	(X, Δ)	$h \le \Delta/M'$
above 2	$\Delta \ge \Delta_c X$	(X, q)	$h \le q/K$
below 1	$-\Delta > M'h$, $-\Delta \le E_c X$	$(X, -\Delta)$	$h \le -\Delta/M'$
below 2	$-\Delta \ge E_c X$	(X, q)	$h \le q/K$

The wedge substitutions are $h = \eta X$ in the near zone; $h = \eta\Delta$ and $\Delta = \zeta X$ with $\zeta \in [0, \Delta_c]$ in above 1; $h = \eta(-\Delta)$ and $-\Delta = \zeta X$ with $\zeta \in [0, E_c]$ in below 1; and $h = \eta q$, $q = \zeta X$ with $\zeta \in [0, 1]$ in above 2 and below 2.

Lemma 6.3. *These zones cover the far-below zone $\{X > Mh,\ q \le X,\ h \le q/K\}$.*

Proof. Let $h \le q/K$. If $|\Delta| \le M'h$, then $q \le \lambda X + M'h$ and $Kh \le q$ give $h \le \lambda X/(K - M')$: near zone. If $\Delta > M'h$: when $\Delta \le \Delta_c X$ we have $\Delta/M' \le (\lambda X + \Delta)/K = q/K$, and $h < \Delta/M'$ (above 1); when $\Delta \ge \Delta_c X$ the original slab is kept (above 2). The case $-\Delta > M'h$ is the same with $E_c = M'\lambda/(K + M')$, the value where $-\Delta/M' = (\lambda X + \Delta)/K$. In each zone the wedge substitutions hold: $h \le \Delta/M'$ with $\Delta \le \Delta_c X$ gives $h = \eta\Delta$, $\eta \in [0, 1/M']$, $\Delta = \zeta X$, $\zeta \in [0, \Delta_c]$, and similarly for the others. □

7. Implementation and soundness rules

The program `ttbox` (Python, about 1,000 lines: `ttpoly.py`, `ttbox.py`, `ttarea.py`, `run_tt.py`, `cov_check.py`) follows these rules.

(1) *Exact data.* Coordinates, weights, chart polynomials, limit values, polygon splits, the values of T at vertices, and all coverage comparisons are exact rationals (Python `Fraction`).
(2) *Enclosures.* Floating-point intervals are used only for the sign tests and derivative bounds of Sections 5. Rational coefficients are converted with one-ulp widening; every operation rounds the lower end down and the upper end up (`nextafter`); polynomial evaluation groups terms by powers of h to limit dependency loss; division by an interval containing 0 raises an error.
(3) *Only positive quantities are divided out*: powers of h (Lemma 4.1), and in the wedge tests powers of the wedge variable, which is positive on the region of the substitution.
(4) *Segment filter.* A segment is skipped on a cell only if its bounding box misses the bounding box of all pose centers of the cell expanded by $3/4$, which exceeds the circumradius $\sqrt{2}/2$ of a unit square; the comparison is exact. (It could only make the bound weaker: every weight is positive.)
(5) *No silent loss.* Every cell ends certified, split into pieces that cover it, or reported as refused (size floor, failed split, or cell cap). A job prints `ALL-OK` only if every chart and every corner zone has no refused cell; a run restricted to some charts now ends `PARTIAL-...` (the two reruns in `out_fix/` predate this rule and end `ALL-OK`; the job check of Section 9 reads the outputs chart by chart, so this does not matter).
(6) *Safety checks* are explicit exceptions, not assertions, so they cannot be disabled.

8. Computation

The poses of Theorem 2.1 are split into jobs, one per base line or strip, depth and tilt sign: 308 jobs with $u_0 = 1/4000$ (`make_jobs.py`) and 15 with $u_0 = 1/2000$ (`run_u2000.sh`). Each job runs every chart of its line or strip.

- 306 of the 308 main jobs were certified directly. The two jobs at $d_0 = 7/2$, $y_b = 3.5$, $\sigma = \pm 1$ each had 11 refused cells in one wedge chart (W_t^+ for $\sigma = -1$, W_d^+ for $\sigma = +1$), at the wedge corners $p^* = -1/5$ ($\rho = 2^{-27}$) and $p^* = 0$ ($\rho = 2^{-26}$), each with a ray $\lambda = 1/2$. They were re-run with the zones of Section 6: all zones certified (center 31 and 15 cells, ray zones 25 and 27 cells).
- In the job $d_0 = 5/2$, $\sigma = -1$, $y_b = 4.5$, $u_0 = 1/2000$, the chart C_u (89,800 cells) has four corner points, $(1, -\frac{4}{5})$, $(1, \frac{2}{5})$, $(1, -\frac{2}{5})$, $(1, -\frac{3}{5})$, all re-covered automatically (544 zone cells).
- In total about 2.75 million cells were certified, with no refused cell left in any job.

9. Coverage of the remaining boxes

Proposition 9.1. *Every sub-box that* `zmx2` *leaves uncertified in the* 75 *tight-tile root boxes consists of poses covered by Theorem 2.1 or of untilted poses.*

Proof. `zmx2` was rerun on these root boxes with the settings of the main sweep and with all uncertified sub-boxes printed: 1,255,023 boxes (x, y, u) with exact rational ends. A box near $x = d_0$ consists of left-wall poses with $\sigma = +1$. A box near $y = d_0$ is mapped to left-wall poses by $(x, y) \mapsto (y, x)$, which preserves μ and reverses orientation, so $\sigma = -1$ and the height is x. The script `cov_check.py` verifies in exact arithmetic that every box has $u \le u_0$, offset at most $\delta_{\max}$ from d_0, and heights in $I(d_0)$, or in a range of Table 1 extended by $1/500$ at both ends (the end lines' charts reach that far) with $u \le 1/2000$. Points with $u = 0$ are untilted. It also re-reads all

323 job outputs and requires, for each job, the complete list of expected charts, each run with the stated u_0 and K and with no refused cell; a rerun overrides exactly the charts it reports, and appended hand-made supplements are rejected. Result: all 323 jobs pass, and all 1,255,023 boxes lie in certified regions. □

10. Controls

The controls are not part of the proof; they test that the checker refuses what it must. Each one runs a *modified* measure on a region where μ itself is certified.

- All weights multiplied by 0.99: refused ($G-1=-0.01$).
- The segments on the line $x=2$ weakened by 1%: refused ($G-1=-5.7\cdot10^{-4}$).
- *Same limit values, smaller growth.* One segment ($x=1.9$, $y\in[5.0,5.1]$, weight 0.0142), which touches the tile centered at $(3/2,9/2)$ only at a point, is moved to $y\in[5.1,5.2]$. No limit value changes, but the pose with center $(3/2+h,\ 9/2+h)$ and $u=h$ has exact mass $1-1.34\cdot10^{-7}$ at $h=10^{-6}$ under the modified measure (and $1+1.22\cdot10^{-7}$ under μ). On the C_u face region $[\frac{3}{4},\frac{5}{4}]^2$ around $(1,1)$, the checker certifies μ with no refused cell (155 cells processed) and refuses 1,190 of the 6,079 cells it processes for the modified measure, six of them containing $(1,1)$.

11. Independent review

An independent review (a separate agent, with no part in writing the code) read this proof and the program adversarially. It found no flaw in the mathematics or in the certifying code. Its checks:

- 44,000 exact point tests of the interval routines, including adversarial coefficients and degenerate boxes: no enclosure failure.
- Exact rational recomputation of $\mu(P)-1$, with all 13,244 segments and the exact area of $P\cap U$, at every vertex of every certified cell in sample regions (C_u at $d_0=3/2$; the $d_0=7/2$ wedge charts with the Lebesgue term; every corner and ray zone of the two re-run jobs): $T\le\mu(P)-1$ everywhere, with margins down to 10^{-19}.
- Exact random and boundary sampling of Lemmas 4.2, 6.1, 6.2, 6.3: no pose missed.

Its two substantive findings concerned bookkeeping in the coverage step (the check of job outputs did not verify the chart lists and parameters, and a hand-made supplement was accepted). Both were fixed as described in Proposition 9.1, and the affected job was re-run.

12. Reproducing the computation

From the folder `tight_tile` of the release (the program `ttbox`):

- `NP=8 bash run_all2.sh` runs the 308 main jobs into `out/`;
- `bash run_u2000.sh` runs the 15 jobs with $u_0=1/2000$ into `out_u2000/`;
- reruns with the corner zones are in `out_fix/` and `out_u2000_fix/` (any job can be re-run with `python3 run_tt.py COVER d0 sign line y_b`);
- `bash run_cov.sh` dumps the uncertified boxes of the 75 roots into `cov/`;
- `python3 cov_check.py cov` prints `COVERAGE-OK` if every check passes.

Most jobs take minutes; the C_u charts of some lines take hours (the longest about 15 hours of one core).

References

[1] M. Khalili and S. Gharabaghi, $k^2 - 5$ unit squares need a square of side k for every $k \geq 7$ (2026).
[2] E. Daniel, square-packing: certificates, checkers and Lean proofs, commit `659ee62`, https://github.com/evand/square-packing (2026).

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